21 · Final Consolidated Report

VELOOP Rewards · Data Analyst Internship · Task 01 · Product Analytics, Data Readiness & Data Management Blueprint · 2026-10-01

> **SIMULATED DATA NOTICE.** Every number below that describes data comes from fictional datasets (supplied by VELOOP for this task, or generated by the analyst and labelled "SIMULATED DATA — CREATED FOR ANALYTICAL TESTING"). None of it describes actual VELOOP users, earnings, withdrawals, revenue, retention, conversion or performance.

What was investigated

The work had three parts. First, what VELOOP will need to measure, based on the brief and 17 recorded sources including the official help center and privacy policy. Second, what the supplied dataset can and cannot support. Third, which collection, structure, quality and governance controls VELOOP should put in place before production reporting starts.

What was discovered

1. **The supplied dataset contains only Users data.** It has 1,000 clean rows and 1,008 audit rows. The seven other datasets the brief describes (Task Activity, Rewards, Wallet, Withdrawals, Referrals, Feature Events, Messy Data) were not supplied. They were generated under section 28, labelled as simulated, anchored to the supplied user IDs, and documented.
2. **The supplied structure is a snapshot, not a history.** Of 14 sampled KPIs, 9 cannot be computed from it and 4 only partially. No engagement, money or referral KPI can be calculated over time.
3. **The "clean" sheet still has logical defects.** 91 users have sessions but zero active days. 30 have a login but zero sessions. 34 are opted into marketing while consent is pending. Activity flags rely on an undocumented reference date (reverse-engineered: 2026-09-15 00:00). The messy sheet's planted defects were all found by rules (12/12 cells, 8/8 duplicate rows), and 90 of 90 defects injected into the generated Messy_Data were detected (0 false positives).
4. **Referrals cannot be measured with the supplied structure.** Referred users have no link to their referrer.
5. **The analytics dataset contains personal data** (name, email, exact age, gender, city) that no analysis needs.

6. **Public sources conflict.** Homepage figures (1.57M+ users, crypto payouts, staking) do not match the brief or the help center. They were recorded as unverified and not used.

What is currently unknown

There are 27 recorded unknowns (deliverable 19). The highest-risk ones: whether any events are emitted today (A-001); the timestamp timezone (A-004); whether a wallet ledger and idempotency keys exist (A-005, A-006); currency conversion rates (A-010); real withdrawal statuses and reasons (A-011); partner callbacks (A-014); the policy for reversals after withdrawal (A-020).

What VELOOP should implement

- A **46-event tracking plan** using one `object_action` naming standard. Money and status events are server-side, and each has an `event_id`.
- A **17-entity logical model** with an append-only wallet ledger, separate activity and reward records, and status histories.
- A **181-field data dictionary** and **53 KPI definitions** (no targets).
- **17 quality rules**, each placed at the earliest layer that can stop the defect.
- A two-stage, low-cost pipeline, a pseudonymous access model, a change log, and a 20-item backlog.

Major risks

- **Duplicate or unreconcilable money records** if transaction IDs and idempotency keys are missing. This is critical because it affects users' balances directly.
- **Wrong daily KPIs** if the timezone is not fixed (the daily-reward window is IST).
- **Misleading reporting** if login-based flags, homepage figures or simulated data are reported as performance.
- **Privacy exposure** from personal data in analytics extracts, and consent conflicts in the marketing audience.
- **Partner dependency:** ad, survey and offer outcomes are decided outside VELOOP and arrive late or not at all.

Major priorities

1. Confirm what exists today (BL-01).

2. Stable IDs, ledger, reward states, withdrawal history and UTC timestamps (BL-02 to BL-06).
3. Server-side critical events and quality checks (BL-07 to BL-11).
4. Remove personal data and fix consent (BL-12). Publish conversion rates (BL-13).
5. Build dashboards only after production data passes the checks (BL-18).

Readiness today: **9/40** (deliverable 18). This low score reflects the evidence available to the analyst, not a judgement on the product.

Deliverable index

no	deliverable	document	spreadsheet
01	Executive Summary	01_Executive_Summary.md	
02	Product Data Audit	02_Product_Data_Audit.md	
03	Data Coverage Matrix	03_Data_Coverage_Matrix.md	03_Data_Coverage_Matrix.xlsx
04	User Analytics Specification	04_User_Analytics_Specification.md	04_User_Analytics_Specification.xlsx
05	Earning & Reward Analytics Specification	05_Earning_Reward_Analytics_Specification.md	
06	Wallet & Withdrawal Analytics Specification	06_Wallet_Withdrawal_Analytics_Specification.md	
07	Referral & Engagement Analytics	07_Referral_Engagement_Analytics.md	
08	Event Tracking Plan	08_Event_Tracking_Plan.md	08_Event_Tracking_Plan.xlsx
09	Data Dictionary	09_Data_Dictionary.md	09_Data_Dictionary.xlsx
10	Logical Data Model	10_Logical_Data_Model.md	10_Logical_Data_Model.xlsx
11	KPI Catalogue	11_KPI_Catalogue.md	11_KPI_Catalogue.xlsx
12	Simulated Data Profiling	12_Simulated_Data_Profiling.md	12_Simulated_Data_Profiling.xlsx
13	Data Quality Audit	13_Data_Quality_Audit.md	13_Data_Quality_Audit.xlsx
14	Dashboard Blueprint	14_Dashboard_Blueprint.md	14_Dashboard_Blueprint.xlsx
15	Developer Analytics Handoff	15_Developer_Analytics_Handoff.md	
16	Data Pipeline Proposal	16_Data_Pipeline_Proposal.md	
17	Privacy & Governance Proposal (incl. Q change management)	17_Privacy_Governance_Proposal.md	17_Privacy_Governance_and_Change_Log.xlsx
18	Analytics Readiness Assessment	18_Analytics_Readiness_Assessment.md	18_Analytics_Readiness_Assessment.xlsx

no	deliverable	document	spreadsheet
19	Assumption & Verification Register (+ Source Register)	19_Assumption_Verification_Register.md	19_Assumption_Verification_Register.xlsx
20	Implementation Backlog	20_Implementation_Backlog.md	20_Implementation_Backlog.xlsx
21	Final Consolidated Report	21_Final_Consolidated_Report.md	

Definition of Done (section 32)

criterion	where it is met
Product data areas have been mapped	03, 02
Major analytical events have been defined	08, 15
User analytics requirements are documented	04
Reward analytics requirements are documented	05
Wallet/withdrawal analytics are documented	06
Referral analytics are documented	07
Event naming is standardized	08 (validated by scripts/validate.py)
Data dictionary is complete	09
Logical data model is complete	10
KPIs have reproducible definitions	11
Simulated data has been profiled	12
Data quality problems have been identified	13
Data quality controls have been proposed	13 (Work Package L)
Dashboard requirements are documented	14
Engineering requirements are specific and testable	15 (acceptance tests)
Data pipeline is documented	16
Governance/access requirements are documented	17
Unknowns are explicitly recorded	19
Analytics gaps are prioritized	18, 03
Final implementation backlog exists	20
All simulated data is clearly labelled	data_environment column on every generated row; banners on every data deliverable

criterion	where it is met
No fictional result is represented as an actual VELOOP result	Banners; validate.py scans documents for unqualified claims
Final documents are internally consistent	All documents generated from one spec; validate.py cross-checks events, fields, KPIs, model and counts

Integrity rules (section 34): how each was respected

Rule	How
1–3 No fabricated production data / simulated not presented as real / no fake claims	All data findings carry the simulated banner and are phrased "in the simulated dataset"; homepage figures are excluded (A-021)
4 No invented security mechanisms	Only reason categories; no thresholds or detection rules
5 No credentials requested	None requested or stored; never-expose list in 17
6 No proposed event claimed to exist	Every event: status Proposed, implementation "Not verified"
7–8 No hidden uncertainty / assumptions not turned into facts	27 assumptions recorded; Confirmed only for facts with an official source ID
9 No unnecessary personal fields	Name, email, exact age, gender and city excluded from the model
10 Accuracy over appearance	Generated data includes failures, rejections and reversals; it surfaced a negative-balance policy gap rather than hiding it
11 Traceability	Every number generated by scripts from named datasets; every fact has a source ID
12 No silent mixing	`data_environment` field and QC-16

Reproducing this work

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python scripts/fetch_data.py                # downloads the supplied workbook
cd scripts && ../.venv/bin/python run_all.py          # generate, profile, audit, build, va
```